SECTION 416

MANUFACTURE AND INSTALLATION OF POST-TENSIONED CONCRETE MEMBERS

DESCRIPTION

416.01 Scope – This Section covers the manufacture and installation of post-tensioned concrete members.

416.02 General Requirements

416.02.01 The requirements of CAN/CSA-S6-00 and DBSS 415 shall apply unless specified otherwise herein or in the Contract.

For items not covered in CAN/CSA-S6-00, DBSS 415 or the Contract, then CSA A23.1, Subsection 6.8, shall apply.

416.02.02 Three weeks before post-tensioning, the Contractor shall submit to the Designer and Ministry Representative for review, complete details of the proposed procedure, including jacking sequence and loads, procedure for testing grout cubes, and a list of equipment. This procedure shall be prepared and sealed by a professional engineer registered with the APEGBC.

MATERIALS

416.11 Anchorages - Anchorages tested in an unbonded state shall develop at least 95% of the minimum specified ultimate strength of the prestressing tendons, without exceeding anticipated set and without slippage. Test results or certificates meeting these requirements shall be supplied by the Contractor.

Anchorages shall be able to sustain the loads applied to them without slippage, distortion or other changes that will permit loss of prestress.

The anchorages shall be of such a nature that they will not kink, neck down or otherwise damage the prestressing steel.

416.12 Ducts - The ducts shall be provided with vents at all high points and at the anchorages, so as to facilitate grouting. Drains may be provided at low points. In any case, in cold weather, care shall be taken to prevent water from collecting and freezing in the ducts. The locations of the vents and drains shall be as shown on the Design.

Cored and external ducts shall be watertight when tested under an internal water pressure of 0.35 MPa.

The vents and drains shall be 13 mm minimum diameter flexible tubes. The connections to the ducts shall be made with metallic structural fasteners.

416.12.01 Internal Tendons - The sheaths for post-tensioning ducts shall be made of galvanized steel or plastic. Galvanizing shall meet the requirements of CAN/CSA G164-M. Plastic sheaths shall meet the requirements of DBSS 416.12.02. The sheaths shall be corrugated, and chemically non-reactive with concrete, tendons, or grout. The inside cross-sectional area of the sheaths shall be at least two times the area of the prestressing steel.

Rigid steel sheaths shall have a wall thickness of at least 0.60 mm and shall be capable of bending to an inside radius of 9 m without distress. Semi-rigid steel sheaths shall have a wall thickness of at least 0.25 mm and shall be capable of bending to an inside radius of 3.50 m without distress.

The ducts shall be spliced with telescopic sleeves. The joint lines between the ducts and the sleeves shall be wrapped with tape or heat-shrink material to make the connections mortar-tight.

416.12.02 External Tendons - Unless otherwise specified, sheaths for external post-tensioning shall be made of plastic. Plastic sheaths, including their splices shall be made of high density polyethylene conforming to ASTM Standard D 3350, cell classification 334420C. The polyethylene sheath shall be manufactured in accordance with ASTM Standard D 2239.

Plastic sheaths shall not be used when the radius of curvature of the tendon is less than 10 m. Sheaths and their splices for external postensioning shall be smooth, seamless and capable of withstanding a grouting pressure of at least 1 MPa.

416.13 Condition - Tendons, ducts, anchor assemblies and other accessories shall be free of grease, oil and other foreign matter. A light coat of rust will be permissible provided any loose rust is removed and the surface of the steel is not pitted.

416.14 Epoxy - Epoxy applied to match-cast segment ends shall be as specified by AASHTO Standard Specifications for Highway Bridges (2002), 17th Edition, Division II, Section 8.13.7.

416.15 Grout –Cement**,** water and admixtures for grout shall conform to the requirements of CSA Standard A23.1, Clause 4.2. Unless otherwise specified, grout for post-tensioning shall have a compressive strength of at least 20 MPa at 7 days and 35 MPa at 28 days. The Contractor shall propose a design mix for the grout and submit it for review by the Designer and Ministry Representative before grouting may begin.

The Contractor shall supply all materials for the grout.

The grout shall have a maximum water/cement ratio of 0.45. The grout shall consist of a mixture of Type GU (general use) cement, water and an approved admixture to control expansion and pumpability. The admixture shall be an aluminum powder base premixed material. Type HE (high early strength) cement may be substituted for Type GU

during low temperature periods. The Contractor shall determine the dosage of the admixture by using volume change and bleeding tests conducted before grouting commences. The Contractor shall make test grout cubes and deliver them to a testing agency acceptable to the Quality Manager and Ministry Representative.

EQUIPMENT

416.21 Stressing Equipment

416.21.01 Supervision - The Contractor shall provide a supervisor skilled in the use of the system of post-tensioning to be used.

416.21.02 Gauges and Calibration - The tensioning system shall be equipped with gauges that have been calibrated for the specific jacks and controls by an approved testing agency within the last 6 months. The dials of the gauges shall be at least 150 mm in diameter. Before stressing commences, certified calibration curves shall be furnished by the Contractor.

The tensioning system or tensioning devices shall be recalibrated if during the work any gauges give erratic results.

416.22 Grout - The grout shall be mixed in a shear type mixer with sufficient capacity to guarantee a continuous supply of grout to the pump. The accessory equipment to batch all materials shall accurately measure all materials.

The pump shall be a positive-displacement type and shall produce an outlet pressure of at least 1 MPa. The pump shall have seals to prevent introduction of oil, air or other foreign substance into the grout and to prevent loss of grout.

The grouting equipment shall utilize gravity feed to the pump inlet from a hopper attached to and directly over it. The hopper must be kept partially full of grout to prevent air from being drawn into the ducts.

The grouting equipment shall contain a screen having clear openings of 2.5 mm maximum to screen the grout prior to its introduction into the grout pump. A screen opening of 5 mm is satisfactory if a grout with a thixotropic additive is used. This screen shall be readily accessible for inspection and cleaning.

The Contractor shall have on site a pressure gauge with a full-scale of no greater than 2 MPa. The Designer and/or Ministry Representative may request the Contractor to install the gauge in the grout line between the pump outlet and the duct inlets to confirm pumping pressures.

The grouting equipment shall be capable of continuously grouting the largest tendon on the project in 20 minutes.

CONSTRUCTION

416.31 Anchorages - The spacing between anchorages shall be sufficient to allow the operation of the stressing jacks to be unimpeded by the adjacent tendons.

The axis of the tendons shall be in line with the anchorages for a minimum of 400 mm.

The load from the anchoring devices shall be effectively distributed to the concrete.

Anchorages shall be rigidly held in position when the concrete in the section is placed. Anchorages shall hold the tensioned tendons without slip of more than 6 mm; otherwise detensioning will be required.

416.32 Ducts - Vents and drains in the ducts shall be mortar-tight, taped as necessary and shall provide means for injection of grout and for sealing. The ends of vents and drains shall be removed 25 mm below the concrete surfaces after grouting has been completed. If drains are used, they shall be vented downward and shall remain open until grouting commences to permit the removal of any water that may be present in the duct.

416.33 Placement and Protection of the Tendons and Other Accessories - The plan and profile location of the ducts shall be as shown on Design. Ducts shall be securely held at intervals of 1000 mm or less to prevent displacement or flotation during the placement of concrete. They shall be placed with a tolerance of ±6 mm in both the vertical and the horizontal directions. The location of the ducts shall permit the post-tensioning steel to follow the prescribed profiles.

The anchorage assemblies shall be placed so that the bearing surfaces are concentric with and perpendicular to the tendons within ±1°.

Immediately before concrete is placed, the post-tensioning steel profile and alignment shall be checked and corrected where necessary. The ducts, grout fittings, vents and drains shall be inspected and damage shall be repaired to prevent concrete from leaking into the ducts.

The grout fittings and the ducts shall be adequately protected from collapse and other damage. Grout openings, vents and drains must be securely anchored to prevent displacement during concrete placing operations.

The ends of the ducts shall be covered to prevent the entry of water or debris after the installation in the forms.

The ducts shall be blown out with compressed, oil-free air immediately after the completion of the concrete pour. The compressed air shall be used to the extent necessary to break up and remove any mortar in the ducts before it hardens. The ducts shall be blown out clear of any water or debris again prior to the placing of the tendons.

Duct ends at match-cast segment ends shall be fitted with gaskets or other sealing devices to prevent infiltration of epoxy during assembly. A uniform pressure of approximately 0.3 MPa shall be applied to the joint during

the epoxy curing period. The pressure may be supplied by stressing a cable to a predetermined level, or by other means.

Segmental section joints of 10 mm to 70 mm wide shall be filled with a sand-cement mortar. Joints over 70 mm wide shall be filled with concrete. The strength of the mortar or concrete used in joints shall be at least equal to the parent concrete. The Contractor shall colour the mortar or concrete to match the parent concrete. The Contractor shall submit a trial mix with a proportion of white Portland cement, if requested, to the Quality Manager and Ministry Representative for review.

All post-tensioning steel shall be installed in the ducts prior to the concreting of the cast-in-place elements.

The concrete shall be vibrated with particular care at each cast-in-place end block (push-through diaphragm) and post-tensioning anchorage location. This will ensure adequate consolidation in the anchorage zones.

The prestressing steel in each tendon shall be pulled immediately after concreting, to ensure that it is free to move and is not bonded by concrete or grout leakage.

416.34 Stressing - The structure shall be stressed only after the concrete has reached its transfer strength as shown on the Design. No tendons shall be stressed until authorized by the Designer and Ministry Representative.

Tendons shall be stressed from the end as shown on the Design.

The supports under the segmental sections shall be able to accommodate horizontal movements caused by the stressing.

The multiple strands composing a tendon in a common duct shall be tensioned simultaneously. Tendons shall be tensioned in sequence as specified in the reviewed procedure. The tendon force at lock-off shall be as shown in the procedure.

Before commencing the stressing operation an initial tensioning force shall be applied to each tendon to equalize stresses and to eliminate slack. This force shall not exceed 10% of maximum jacking force.

After the initial tension forces have been applied, reference points shall be established for each tendon from which elongation by final tensioning forces will be measured.

The specified tensioning force shall be maintained steady in the strands until they are anchored. The sequence of tensioning shall be such that no undue tension stresses occur in the concrete near the ends of the members.

The tensioning force shall be determined by the calibrated gauge and by the elongation of the strands. When there is a difference of over 5% between the two means of calculation, the cause of the discrepancy shall be ascertained and corrected.

Tendons shall not be subjected to jacking forces in excess of 85% of their specified ultimate tensile strength. Any tendon subjected to such forces will be rejected.

The required tendon elongation and jacking force shall be computed, based on the required prestressing force and the estimated prestress losses.

The safety of the stressing procedure shall be in accordance with Workers Compensation Act, Occupational Health and Safety Regulations, BC.

Stressing records shall be kept by the Contractor. Stressing records shall clearly identify the tendons that were stressed, the sequence, the final jacking loads, the tendon elongations, the times, the temperatures and the dates. Stressing records shall be signed by the person supervising the stressing and submitted to the Quality Manager and if requested, the Ministry Representative.

416.35 Grouting - The temperature of grout shall be controlled by water temperature. Grout shall not be warmer than 30°C or colder than 5°C at the pump. The grout shall be mixed for a minimum of 2 minutes. The materials shall be charged in this order: water, cement and admixture. The admixture shall be added in accordance with the manufacturer's instructions.

The grout shall be mixed to a uniform consistency of thick cream without lumps or undispersed cement. The grout shall be agitated continuously and shall be recirculated when not injecting from the line.

Water shall not be added to the grout to increase the flowability that was decreased by delayed use of the grout. The grout shall be discarded when flowability has so decreased it cannot be pumped satisfactorily.

The time between the tensioning and grouting of a tendon shall not exceed 4 days.

Cored ducts with concrete walls shall be flushed with clean water to ensure that concrete is thoroughly wetted prior to grout injection. The ducts for external tendons shall be checked for leakage and blockage by flushing with clean water under pressure followed by oil-free compressed air immediately prior to grouting. Metal sheaths for internal ducts shall be flushed to clean out foreign materials, or oil free compressed air shall be used to check for blockages.

All grout and high point vent openings shall be open when grouting starts. Grout shall be allowed to flow from the first vent until any residual flushing water or entrapped air has been removed. The vent shall be capped or otherwise closed at that time. The remaining vents shall be closed in sequence in the same manner.

The pumping pressure at the tendon inlets shall be at least 0.7 MPa and shall not exceed 1.5 MPa.

If the pumping pressure exceeds the maximum allowable,

grout may be injected at any grouted vent, as long as a one-way flow of grout is maintained. If this procedure is used the vents that are to be used for injection shall be fitted with positive shutoffs.

The grout shall be pumped through the ducts and continuously wasted at the outlet pipes until no visible slugs of water or air are ejected. The efflux rate of the ejected grout shall not be less than the influx rate of the injected grout.

To ensure that the tendons remain filled with grout, the outlets shall be closed and the pressure held for 10 seconds. The grouting ends shall then be plugged. Plugs, caps or valves shall not be removed or opened until the grout has set. Alternatively, the outlets and inlets shall be kept under a 1000 mm head of hydrostatic pressure. The 1000 mm grout head shall be maintained until the grout has hardened.

The grout shall be immediately flushed out of the ducts with water when one-way flow of grout cannot be maintained. Standby water flushing equipment, with its own source of power, shall be available in addition to the grouting equipment. It shall be capable of developing a pressure of 2 MPa, and sufficient capacity to flush out any partially grouted enclosures. After flushing, the ducts shall be blown out with oil-free compressed air and then checked for blockages. The grouting procedure shall be repeated if there are no blockages.

The ducts shall be completely filled with grout. The presence of ungrouted lengths of duct shall be cause for nonacceptance of the members. Members with partially ungrouted duct(s) shall be replaced or repaired to the satisfaction of the Designer and the Ministry Representative.

Great care shall be taken to prevent blockages of the ducts due to grouting of nearby ducts or other causes.

The grouting shall not be done when air temperatures are above 32°C or below 5°C, nor when freezing temperatures are predicted within the next 24 hours.

The temperature of the concrete shall be 2°C or higher from the time of grouting until site-cured cubes of grout reach a compressive strength of 6 MPa.

The grouted ducts may be subject to gamma ray radiography inspection by the Ministry to ensure that all ducts are fully grouted. The cost of the inspection will be borne by the Ministry if the inspection indicates that ducts are completely filled with grout. Otherwise, the costs of inspection and the cost of reinspection shall borne by the Contractor.

416.36 Cropping - The cropping of post-tensioning strand shall be by a cold-cutting procedure. It shall not take place until authorized by the Designer and Ministry Representative. Under no circumstances shall heat be applied to post-tensioning strand from cropping or other operations. The cropped strand shall be provided with a minimum concrete cover of 25 mm.

416.37 Anchorage Recesses - The anchorage recesses shall be cleaned and sandblasted after completion of post-tensioning and grouting. The surfaces shall be coated with an epoxy bonding agent conforming to ASTM C 881,Type V, Grade 2. The Class of bonding agent shall be appropriate for the temperature at time of application. The bonding agent shall be applied in accordance with the manufacturer's applications. Wire mesh shall be placed before concreting in recesses larger than 300 mm square.

The Contractor shall fill the recesses flush with adjoining surfaces. All parts of anchorages shall have a minimum cover of 50 mm unless otherwise approved by the Ministry Representative. All recesses exposed to view shall be filled with concrete coloured to match the parent concrete. If requested, the Contractor shall submit a trial mix with a proportion of white Portland cement to the Quality Manager and Ministry Representative for review.

SECTION 418

CONCRETE BRIDGE DECK SEALANTS

SILANE SURFACE TREATMENT OF NEW PORTLAND CEMENT CONCRETE

418.01 Scope - This Section describes the treatment of new concrete bridge surfaces with a silane protective solution. The Specification describes surface preparation and application.

Surfaces to be treated shall be as specified on the Design and in the Contract.

418.02 Silane Solution - The silane solution shall be a minimum 40% solution of alkyl trialkoxy silane (isobutyl trimethoxy silane or n-octyl triethoxy silane) dissolved in a maximum of 60% anhydrous alcohol.

418.02.01 Delivery and Storage of Silane Solution - The silane solution shall be delivered to the Site in unopened, sealed containers with original seals intact, also with a manufacturer's label identifying the product (including a batch number). The containers holding silane solution shall be stored in dry, enclosed structures to protect the contents from misuse, weather and excessive heat.

418.02.02 Quality Assurance - Only silane solutions listed in the Recognized Products List shall be permitted for use on the job.

The Recognized Products List may be found on-line at: http://www.th.gov.bc.ca/publications/eng_publications/geotech/Recognized_Products_Book.pdf

Certification shall be required for each batch shipment of silane solution, prior to its use.

418.03 Contractor's Qualification - The applicator shall have a minimum of three years of successful and documented experience in application of liquid water repellents to concrete or masonry substrates, or else the applicator must provide documentation certifying that training approved by the manufacturer for applying silane solution was received.

418.04 Site Conditions

418.04.01 Substrate Condition - Prior to silane treatment:

- New concrete shall be cured for a minimum of 28 days.
- All concrete surfaces shall be fully dry.

418.04.02 Weather Limitations - Application of silane solution shall be undertaken only during periods when rain or snow has not occurred during the previous two days and no rain or snow is forecast by the weather office for the next two days.

Application of silane solution shall proceed only when the concrete surface temperature is above 4°C and below 32°C, and windspeed is less than 24 km/hr.

418.05 Surface Preparation - All concrete surfaces prepared for treatment shall be thoroughly cleaned prior to application of the silane solution. The method of cleaning shall be capable of removing loose concrete, curing compounds, laitance, dirt, dust, salt, oil, asphalt or other foreign materials. The cleaning process shall expose concrete pores and capillaries to readily accommodate silane penetration.

The cleaning process shall not damage the concrete surface, remove or alter the existing surface finish, or expose the coarse aggregate of the concrete. The method of cleaning shall be performed in such a manner as to provide a uniform appearing surface colour.

418.05.01 Pre-Surface Cleaning - If necessary, pre-surface cleaning preparations may include the use of solvent and hand tools to remove materials detrimental to the silane treatment of the concrete surface.

418.05.02 Surface Cleaning - Suitable methods to clean the concrete surface are:

- Sand blasting
- Steel shotblasting
- Hydro blasting
- Steam cleaning
- Water pressure washing using a minimum of 3.5 MPa (500 psi) nozzle pressure.

The cleaning method used shall be capable of cleaning concrete surfaces, as specified.

418.06 Silane Application - Silane application shall proceed only if the condition of the cleaned concrete surface is approved by the Quality Manager and Ministry Representative.

418.06.01 Surface Condition - All concrete surfaces shall be fully dry prior to silane solution application. Concrete surfaces previously steam cleaned or water washed must be allowed to dry prior to application of silane solution for:

- Minimum one day during windy or sunny days when air temperatures are over 20°C.
- Minimum two days during calm or cold days when air temperatures are between 4°C to 20°C.

418.06.02 Silane Solution - The silane solution shall be used as supplied by the manufacturer. It shall not be diluted or altered in any way. Water shall not be allowed to contaminate the silane solution.

418.06.03 Application Equipment - The silane solution shall be sprayed on the concrete surface by using low pressure 0.1 MPa - 0.2 MPa (15 - 30 psi) positive displacement airless spray equipment. Suitable equipment is a pressurized garden sprayer or drum mounted pump. Large deck areas over 400 m^2 shall be sprayed using a spraybar type unit.

All spray equipment shall be fitted with fan spray nozzles and the spray adjusted to a "wet spray" condition (i.e., minimum atomization of silane solution).

i) Cleaning of Application Equipment - All application equipment must be clean and dry prior to use. If equipment is wet, it shall be rinsed with a small amount of methyl hydrate (methanol) prior to commencement of the actual application.

418.06.04 Rate of Application

i) Surfaces Subject to Traffic Wear - Bridge decks and other surfaces subject to traffic wear shall be treated with silane solution at a rate of 0.40 ℓ/m^2.

ii) Surfaces not Subject to Traffic Wear - Surfaces not subject to traffic wear (parapets, pier caps, sidewalks, etc.) shall be treated with silane solution at a rate of 0.23 ℓ/m^2.

418.06.05 Method of Application - Silane solution shall be applied at a coverage rate according to this Specification. Horizontal surfaces shall be uniformly sprayed to saturate/flood the surface. Vertical surfaces should be treated from the bottom up and uniformly sprayed to saturate the surface.

If it is not possible to complete the silane treatment in one application, the boundary between the treated and untreated areas shall be clearly marked.

418.06.06 Curing - Following the application, the surface of silane treated concrete shall be cured by mist spraying with water after:

- 4-6 hrs, during days with temperatures between 25^oC - 32^oC
- 6-12 hrs, during days with temperatures between 15^oC - 25^oC
- 12-24 hrs, during days with temperatures between 4^oC - 15^oC

The traffic may be allowed on silane treated concrete one hour after the water wetting operation.

418.07 Safety and Health Hazards - Personnel exposed to silane solution and methyl hydrate shall wear personal protective equipment such as coveralls, goggles, rubber gloves and proper respirators.

No smoking, open flames, sources of heat or ignition shall be permitted during the construction operation.

Special precautions must be taken during the construction operation to prevent any over spray or spillage of materials on plants and vegetation or into fish-habitat water bodies, (Reference DBSS 165, Protection of the Environment).

Manufacturer's material safety data and handling instructions shall be followed.

SECTION 419

CONCRETE BRIDGE DECK WATERPROOFING SYSTEM

PREFABRICATED MEMBRANE

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

419.01 Scope - This Section describes the protection of concrete bridge decks with rubberized asphalt prefabricated membrane overlaid with hot mix asphalt pavement. The Section describes materials, surface preparation and application.

NOTE: This Section is not applicable to existing concrete with high levels of chloride contamination. Membrane and pavement overlay shall not be installed on concrete bridge decks contaminated with acid-soluble chloride ion (Cl^-) in excess of 0.10% by mass of cement.

MATERIALS

419.11 General - The membrane and accessory materials shall be supplied as a package containing:

a) full-width membrane, self-adhering internally reinforced sheet of rubberized asphalt. Membrane shall have a minimum thickness of 1.6 mm (65 mils) and shall be supplied in rolls in width between 900 mm and 1500 mm (3' to 5').

b) detailing strip of membrane, self adhering internally reinforced sheet of rubberized asphalt. Membrane shall have thickness between 1.50 and 1.75 mm (60 to 70 mils) and shall be supplied in rolls in width between 200 and 305 mm (8" to 12").

c) primer, rubberized asphalt dispersed in organic solvent with minimum solids of 45% (by mass), and maximum drying time of one hour.

d) mastic, single component material, consisting of rubberized asphalt, organic solvent and fillers, supplied in caulking tubes.

419.12 Delivery and Storage - Membrane and accessory materials shall be delivered to the Site in unopened packages and containers with manufacturer's label identifying the product and a batch number.

The packages and containers shall be stored in a dry secured place protected from weather and excessive heat. Specific manufacturer's instructions for storage shall be followed.

419.13 Quality Assurance - Only membranes and materials accessories (supplied as a package) from the Recognized Products List prepared by the Ministry will be permitted for use.

Certification shall be provided for each batch (package) shipment, prior to its use.

CONSTRUCTION

419.31 Applicator's Qualification - The applicator must have a minimum of three years of successful and documented experience in application of solid membranes to concrete substrates, or else the applicator must provide documentation certifying that training approved by the manufacturer for applying solid membranes has been received.

419.32 Site Conditions

419.32.01 Substrate Conditions - Prior to application of membrane, new concrete shall be cured for a minimum of 28 days.

419.32.02 Weather Limitations - Application of primer and installation of membrane and pavement overlay shall be undertaken only during stable weather when precipitation has not occurred during the previous 24 hours and is not imminent, and when the minimum ambient and concrete surface temperature is 5^oC and rising.

419.33 Surface Preparation - All concrete surfaces shall be sound, smooth, clean and free from any contamination.

a) All defects in concrete surfaces, cracks, delamination and spalled areas shall be repaired with a suitable patching material. All patching shall be completely cured.

b) All protrusions and sharp edges shall be removed by using hand tools or power tools.

c) New concrete overlays on bridge decks and concrete-filled shear keys between prestressed concrete box stringers shall have smooth finish. Broom finishes are not acceptable.

d) All loose aggregates, laitance dust, soil, oil and other debris and contaminations shall be removed from the concrete surface by one or a combination of the following methods:

i) high pressure clean dry air sweeping

ii) sand blasting

iii) steel shotblasting

iv) hydro blasting using a minimum of 3.5 MPa (500 psi) nozzle pressure

v) steam cleaning.

The method used shall be capable of cleaning concrete surfaces as specified.

419.34 Priming - All concrete deck surfaces that are to receive membrane must be primed.

419.34.01 Concrete Surfaces - All concrete surfaces shall be fully dry prior to application of primer. Concrete surfaces previously steam cleaned or water washed must be allowed to dry prior to application of primer for:

- minimum 24 hours during windy or sunny days when air temperatures are over 20°C.
- minimum 48 hours during calm or cold days when air temperatures are between 5° to 20°C.

419.34.02 Primer Material - Primer shall be used as supplied by the manufacturer. It shall not be diluted or altered in any way.

419.34.03 Primer Application - Primer application shall proceed only if the condition of cleaned and repaired concrete surfaces is approved by the Quality Manager and the Ministry Representative.

All concrete surfaces to be covered with membrane shall be uniformly treated with primer at the rate of 0.15 to 0.25 ℓ/m² depending on the porosity of the substrate surface.

The primer shall be uniformly applied on substrate surface by roller or brush.

419.34.04 Primer Curing - Only as much surface as will be covered by membrane in the workday shall be primed. Primed concrete surfaces not covered by membrane within the workday must be reprimed immediately prior to membrane installation.

All primed areas must be tack free prior to application of the membrane.

419.35 Membrane Application

419.35.01 Detail Work - Extra attention must be given to all detail work. Detail work shall be performed first, prior to installation of the full-width membrane. The membrane shall be installed as called for on the Design and on SS Drawings SP419-01 through SP419-08 and as follows:

i) curb or parapet - inside corners at curbs or parapets shall be covered using a detailing strip of membrane. The uppermost edge of the detailing strip of membrane shall be placed and extended up the face of curb or parapet to 6 mm (1/4") below the finished height of the asphalt pavement overlay. The detailing strip of membrane shall be installed in such a manner to ensure it is smooth and that it adheres strongly to the face of curb or parapet, and bridge deck without any visible defects ("tenting", "fish mouth" or "bubbles"). The first sheet of the full width of membrane shall then be applied as close as possible to the curb or parapet.

ii) bridge deck edge - on bridge decks without curbs or parapets, the full width membrane or detailing strip of membrane shall terminate 6 mm (1/4") from the outer edges of the deck. The membrane shall be heated at the edge of the deck with light flame from a torch or with a hot air gun, followed by pressing the membrane down with a roller.

iii) drains - areas around drains, basins and any other protrusions shall be double covered with membrane. First, flashing pieces of detailing strip of membrane are applied, followed by the application of the full width membrane. The membrane shall be installed in such a way as not to obstruct openings of drains and basins.

iv) weep holes - do not require detailing strip. The membrane around weep holes shall be heated with light flame from a torch or hot air gun, and pressed down to expose the opening of the hole.

v) joints and cracks - all filled cracks wider than 3 mm (1/8"), and all joints with a gap less than 6 mm (1/4"), shall be covered with a detailing strip of membrane prior to application of a full width membrane. Joints wider than 6 mm (1/4"), shall be treated as a standard expansion joint and membrane shall be terminated at expansion joint assembly.

vi) bridge ends - detailing strip of membrane shall be installed over the joint between the existing pavement and the bridge deck. At each end of the bridge deck, a full-width membrane shall extend at least 900 mm (3') over the paved bridge approaches. In the case of new construction, when the existing structure has gravel approaches, the detailing strip of membrane shall be folded over the deck edge and covered with a full width membrane.

419.35.02 Membrane Installation - The full width membrane shall be applied in a longitudinal direction from the low point to the high point so that laps shed water. The

area along the lower edge of the deck shall be covered first.

The full width membrane shall be laid flat, avoiding wrinkles and entrapped air. Each succeeding sheet of membrane shall be applied overlapping the side laps a minimum 65 mm (2 1/2") and the end laps a minimum 150 mm (6"). The entire surface of the membrane, especially laps, and all terminations shall then be rolled with sufficient pressure to affect a bond between the membrane and the concrete surface.

419.35.03 Mastic Application - All membrane terminations shall be sealed with a trowelled application of mastic 12 to 25 mm (1/2" to 1") wide and 1.5 to 3.0 mm (1/16" to 1/8") thick. Mastic shall be applied along the edge of the membrane at the face of the curb or parapet, along both sides of the deck joints, around drains and at termination of the membrane on both ends of bridge approaches.

At the end of each working day, or when application is interrupted by a change in construction scheduling, all membrane terminations shall be sealed with the mastic.

419.35.04 Inspection and Repairs - Prior to paving over the membrane, any defective areas of the membrane shall be repaired to the satisfaction of the Quality Manager and Ministry Representative, including:

- tears, damaged areas and inadequately lapped seams shall be patched;
- fish mouth shall be slit and repaired;
- large blisters shall be punctured and repaired; and
- if water migrates underneath the installed membrane, then the membrane covering the affected area shall be removed and replaced.

419.35.05 Traffic On Membrane - Traffic shall not be permitted on the membrane prior to installation of asphalt overlay with the exception of vehicles and equipment engaged in installing the asphalt overlay. Only vehicles or equipment provided with pneumatic tires or with flat tracks with rubber bonded pads, shall be permitted on the membrane. Any turning or sudden braking on the membrane shall be avoided. Any damage to the membrane resulting from the movement of vehicles or equipment shall be repaired.

419.36 Paving - Asphalt overlay construction shall proceed only when the final condition of the installed waterproofing membrane is accepted by the Quality Manager and Ministry Representative.

Prior to commencing asphalt pavement construction, the weight of equipment, including loaded trucks, paver and rollers, and their maximum combined weight must be accepted by the Designer. Only this accepted equipment and equipment combination shall be permitted on the bridge structure at the time of construction. Asphalt materials, mix design, mix production and hot mix asphalt pavement construction shall meet the requirements of DBSS 501 and DBSS 951.

419.36.01 Asphalt Mix and Pavement Design - Asphalt mix shall meet the requirements of DBSS 501 for Pavement Class I, Medium Mix and appropriate asphalt binder. The asphalt pavement thickness constructed over the membrane shall be not less than 100 mm and shall be constructed as indicated in DBSS 501.

419.36.02 Placing The Asphalt Mix - In addition to standard paving practice:

i) no paving shall proceed over wet membrane;

ii) no primer shall contaminate membrane surface;

iii) paving operation shall commence as soon as possible after the waterproofing membrane is installed;

iv) paving shall proceed in the direction of the end laps in the membrane (i.e., from overlap to underlap);

v) the Contractor shall take due precautions to avoid damage to the waterproofing membrane during the paving operations by the paver and hauling trucks;

vi) the paver shall move at a constant speed avoiding any unnecessary stopping;

vii) hauling trucks shall maintain a steady supply of asphalt mix to the paver;

viii) excessive accumulation of asphalt mix in the paver hopper and auger shall be avoided;

ix) unloading of the asphalt mix from the hauling trucks onto the membrane surface will not be permitted;

x) paver screed burners shall be set in such a way as to avoid any damage to the membrane from excessive heat;

xi) the path in front of the screed side end plate must be kept continuously free from spilled asphalt mix. The bottom of the side end plate shall slide freely, otherwise the entrapped asphalt mix could damage the underlying membrane;

xii) screed adjustments during paving shall be made in such a way as to achieve a uniform thickness of mat, surface texture, smoothness and designed pavement geometrics;

xiii) the bridge deck joints, drains and other openings (except weep holes) shall be protected to avoid intrusion

of asphalt during the paving operation; and

xiv) during a "one lane at a time" construction method, as on bridges partially open to traffic, the constructed pavement mat shall terminate not less than 100 mm (4") from the edge of installed membrane in order to provide a sufficient area for lapping of the next sheet of membrane.

419.36.03 Compaction - In addition to standard compacting practice:

i) drum vibration mode must not be used during compaction.

ii) steel drum shall run as close as possible to the curb or parapet.

iii) compaction along the expansion joints or concrete dams shall be performed transversely.

iv) sudden stops or sharp turns by rollers shall be avoided.

419.36.04 Hand Work - Placing of asphalt mix in areas inaccessible to paving equipment shall be done in accordance with DBSS 501.35.10 and DBSS 501.36.08.

419.37 Protection Of The Environment - The Contractor shall comply with DBSS 165, environmental requirements of Fisheries and Oceans Canada, the Ministry of Environment, Lands and Parks and any related municipal regulations.

Special precautions shall be taken during the construction operation to prevent spillage of materials into water bodies.

419.38 Safety and Health Hazards - Materials used in the construction of membrane and asphalt pavement overlay are flammable, toxic and hot. Personnel exposed to these materials and operations shall wear all appropriate personal protective equipment such as boots, coveralls, goggles, rubber gloves and proper respirators as specified by Workers' Compensation Board. Manufacturer's material, data and handling instructions shall be followed.

SECTION 421

STRUCTURAL STEELWORK

DESCRIPTION

421.01 Scope - This Section covers the supply, fabrication, erection and quality management program of all major structural steel elements.

In general, all shop connections shall be welded, and all field connections high-strength bolted.

Except as otherwise specified herein, steelwork shall be fabricated and erected in accordance with the AASHTO Standard Specifications for Highway Bridges (17th Edition - 2002 including all subsequent interim revisions and updates), Division II, Section 11. Welding and associated work shall be in accordance with the current edition of CSA W59-M - Welded Steel Construction. Fracture control requirements, welding requirements for fracture critical members and welding requirements for primary tension members shall be in accordance with CAN/CSA-S6-00.

421.02 Prefabrication Meeting - The Quality Manager will convene a prefabrication meeting with the Contractor, fabricator and the Ministry Representative to review procedures on quality control, quality assurance, welding procedures, procedures for non-destructive testing, mill certificates and heat numbers, splices, coatings, updated schedule for fabrication, and to confirm the Contract and Design requirements.

421.03 Alternative Details - All details shall, in general, conform to those shown on the Design. Any proposed variation shall be submitted to the Designer in writing with copy to the Ministry Representative, with the reason for the variation. The submission of Drawings of alternative details shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of British Columbia (APEGBC).

If, in the opinion of the Designer, the variation represents a clear improvement, it may be accepted.

If the variation results in a product of equivalent quality, it may be accepted at the discretion of the Designer.

Alternative details must meet the requirements of the Contract.

421.04 Working Drawings - Working drawings shall consist of the following:

- Shop drawings
- Transportation details and
- Erection drawings

Transportation details and erection drawings shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of British Columbia (APEGBC).

When the Contractor is responsible for the design of items that are detailed on the shop drawings, the shop drawings shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of British Columbia (APEGBC).

Working drawings shall be in the same system of units as the Design.

Working drawings shall be on D size sheets approximately 560 mm (22") by 865 mm (34"), (one drawing per sheet). Lettering for notes and dimensions shall be at least 2.5 mm and 4 mm for headings. Drawings shall be legible when half-sized or microfilmed.

421.04.01 Shop Drawings – Shop drawings shall show all information and details needed for the fabrication of the members including, but not limited to, such items as member shapes and dimensions, connection details, material and product standards, mark numbers and general arrangement of member locations, details of attachments, fastener details, weld details, mass of members, special tolerances, special handling instructions, coating details, lifting details and lifting locations.

421.04.02 Transportation Details – Transportation details shall include such items as:

- Description of hauling and handling equipment
- Weight of members
- Length and height of loads
- Location and method of member support
- Details for handling, storing, and loading of members.

421.04.03 Erection Drawings - Erection drawings shall show in detail the method of erection including, but not limited to, the following:

- Erection procedures
- Procedures for off loading of members upon delivery
- Details for temporary storage and support of members on site prior to erection
- Equipment to be used
- Layout or general arrangement drawing showing the layout of the members, equipment positioning, and access roads
- Crane make, model, and capacity charts, boom length(s) , crane placement, and access for transporting of members to crane(s)

- Radii and loads for crane lifts
- Rigging details
- Mass of members , rigging and special installation equipment
- Details for installation and removal of all falsework, temporary supports, temporary bearings, bracing, guys, dead-men, and lifting devices
- Attachments to the bridge members and bridge structure for temporary support and special launching equipment
- Detailed description of sequence of operations
- Details for special installation equipment such as a launching truss, launching nose, head frames, spreader beams and rollers
- Details for installation of members onto the permanent bearings
- Traffic control plan for roadway and rail traffic
- Details for protection of existing utilities affected by the erection procedures
- Layout and details of fall protection and their sequence of installation.

The Contractor shall be responsible for the lateral stability of members and shall design and provide bracing as necessary until completion of the Work.

The erection drawings shall be complete in detail for all anticipated phases and conditions during erection and during the temporary support of members. The Contractor shall submit calculations, upon request, to the Designer and Ministry Representative that demonstrate that specified factored demand/capacity ratios or allowable stresses are not exceeded in members, falsework, temporary bracing and temporary supports and that member capacities and final geometry will be correct. The calculations shall be sealed by the professional engineer that sealed the erection drawings.

Falsework, temporary supports and temporary bracing shall meet the requirements of CSA Standard S269.1 – 1975, “Falsework for Construction Purposes” and shall also meet all the requirements for falsework given in Clauses 20.17 to 20.26 inclusive of the WCB Occupational Health and Safety Regulations.

A professional engineer registered with APEGBC shall be responsible for any field designs and any changes made to the erection procedures. Field designs and changes to the erection procedures must be documented and sealed by the responsible professional engineer and must be available at the Site prior to the affected erection work being carried out.

Immediately before placement of loading on falsework, the Contractor must ensure that the falsework is inspected and a sealed engineering certificate is issued by a professional engineer registered with APEGBC which:

- Indicates the specific areas inspected and
- Certifies that the falsework has been erected in accordance with the latest approved erection drawings and supplementary instructions.

421.04.04 Submittals - The Contractor shall submit to the Designer copies of all working drawings. Copies shall also be provided to the Ministry Representative. These drawings shall be submitted at least fourteen days prior to the fabrication of the work. Any work done or materials ordered prior to the review of the working drawings shall be at the Contractor's risk. The Designer will review the drawings for general compliance with the Design and the Contract.

If modifications to the drawings are required, the Designer will return one set of drawings, marked up, to the Contractor. The Contractor shall re-submit copies of the revised drawings to the Designer for review. Copies shall also be provided to the Ministry Representative.

If no exceptions are taken to the drawings, the Designer will return one set of reviewed drawings to the Contractor. The Contractor shall forward two sets of the reviewed drawings to the Ministry Representative.

Shop drawings will not be reviewed without the transportation details and erection drawings applicable to the members in question.

Erection will not be allowed to proceed without the Designer’s review of the method proposed.

Review of working drawings shall not relieve the Contractor of responsibility for carrying out the Work in accordance with the Contract and the Design.

421.05 Quality Control - The Contractor shall implement a quality control program to meet the Contract requirements.

421.06 Quality Assurance and Audit - The Quality Manager will implement a quality assurance program by auditing the Contractor's quality control program and by inspection at its discretion. The Ministry will implement a quality audit program by auditing the quality control and quality assurance programs implemented by the Contractor and by inspection and testing at its discretion.

The Contractor shall notify the Quality Manager and Ministry Representative at least 14 days before fabrication is to commence. The Contractor shall allow the Quality Manager’s and Ministry's representatives access to all parts of the work, and shall supply such information and assistance as is required. The Contractor shall provide samples of any materials requested by the Quality Manager and Ministry Representative. Inspection by the Quality Manager and audit by the Ministry Representative

shall not relieve the Contractor from obligation to perform the Work in accordance with the Contract and the Design.

The Quality Manager will be required to test welds by non-destructive testing methods, in accordance with CSA W59-M. Fillet welds will be tested by the dry powder magnetic particle method and butt welds by radiography.

The Quality Manager will attempt to schedule non-destructive testing operations so as not to interfere with the progress of the Work.

The extent of non-destructive testing is required to be as follows:

- Butt Welds 100%
- Flange/web fillet welds 100%
- Flange/stiffener fillet welds 100%
- Web/stiffener fillet welds 25%
- Bracing/stiffener fillet welds 25%

The inspection of welds shall be carried out as soon as possible after the completion of welding. The Ministry Representative may carry out additional non-destructive testing as may be deemed to be necessary.

The Contractor shall be prepared to move and support the pieces being inspected. In general, inspection shall be done on the flat and a minimum of 1.25 m of headroom shall be available.

The Quality Manager shall reinspect welds after defects are repaired.

The Quality Manager will inspect bolted connections in accordance with the AASHTO Specifications for Highway Bridges (17th Edition - 2002 including all subsequent interim revisions and updates), Division II, Section 11.

The Quality Manager and Ministry Representative shall be notified of any defects found in the work. No repair shall be made until agreed to by the Designer and Ministry Representative. In the case of minor or routine corrections, approval to proceed may be given verbally by the Designer with notification to the Ministry's Representative. In such cases as repair of major cracks or a revised design to compensate for deficiencies, the means of correction shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of BC (APEGBC). It shall be submitted in writing, with adequate sketches, to the Designer and Ministry Representative for review.

Each main member shall be certified as acceptable by the Quality Manager before it is shipped from the shop. This certification shall not relieve the Contractor of responsibility for subsequent damage or for defects which become apparent before the Work is finally accepted.

421.07 Quality and Details of Welds - The quality and details of welds shall be in accordance with CSA W59-M.

421.08 Design Specifications - CAN/CSA-S6-00 shall be used in the design of alternative details and calculating the effect of stresses incurred in fabrication and erection.

No increase in allowable stresses due to vertical loads shall be used for erection conditions.

MATERIALS

421.11 Materials

421.11.01 Structural Steel - Steel shall conform to the requirements of CAN/CSA-G40.20/G40.21-M, and shall be of the grades called for on the Design. Plates provided from coils shall not be used. Prior to fabrication, the Contractor shall supply to the Quality Manager and the Ministry Representative the manufacturer's mill certificates giving details of all chemical and physical properties of steel to be used in the Work.

Chemical composition of steel to be galvanized shall be in accordance with DBSS 421.42.

Steel shall be supplied free of surface defects and internal discontinuities, with due regard for the end use of the steel in the Work.

Edges of all plates will be subject to inspection by the Quality Manager. Any discontinuities will be examined and may be accepted.

The Quality Manager and Ministry Representative shall be supplied with a record of all observed discontinuities and repair methods.

Repairs to defective plates shall not proceed until review of the proposed repair method by the Designer and Ministry Representative.

421.11.02 Welding Consumables - All electrodes shall match the base metal specified in accordance with CSA W59-M, Table 12.1. The deposited weld metal shall provide strength, ductility, impact toughness and corrosion resistance equivalent to the base metal.

Welding consumables for all processes shall be certified by the Canadian Welding Bureau (CWB) as complying with the requirements of CSA W48.01.

421.11.03 High-Strength Bolts - Unless otherwise specified on the drawings, high-strength bolts, nuts and washers shall conform to the requirements of ASTM A325, and as follows:

a) for painted steelwork, Type 1, hot-dip galvanized to the requirements of CAN/CSA-G164-M.

b) for unpainted weathering steel, Type 3.

c) for painted weathering steel, Type 3 bolts will be acceptable provided that they are painted after installation in accordance with the field painting specifications.

421.11.04 Shear Connector Studs - Shear connector studs shall conform to the requirements of ASTM A108, Grades 1015, 1018 or 1020.

421.12 Material Storage and Care

421.12.01 Steel - Structural material, either plain or fabricated, shall be stored above the ground in an upright and shored position upon platforms, skids, or other supports. It shall be kept free from dirt and other foreign matter, and shall be protected as far as practical from corrosion. Long members shall be supported on skids placed near enough together to prevent overstress from deflection.

Prior to fabrication, all steel shall be marked for identification by heat number and specification by a marking system approved by the Quality Manager.

421.12.02 Welding Consumables - Electrodes and fluxes shall be stored and kept in condition as required by CSA W59, Section 5.2.

Gas for welding shall be stored in marked steel bottles and shall not be subjected to temperatures in excess of 50°C or temperatures of less than 0°C.

421.13 Fabrication - Prior to fabrication, the welders and welding operator's qualifications, shop drawings, welding procedures, mill certificates and welding consumable certificates shall be submitted for the Quality Manager's review.

No fabrication, welding or coating of steelwork shall commence until permission to do so has been received from the Quality Manager.

421.14 Ministry's Plant Office - The Contractor shall provide in the fabrication plant or nearby a suitable lock-up office for the sole use of the Ministry Representative throughout the period of fabrication.

The office shall be at least 2.5 m x 3 m and shall be weathertight and reasonably soundproof, provided with ample window area and ventilation, furnished with a 0.75 m x 1.2 m desk, a 1 m x 2 m drafting table, a two-drawer filing cabinet, two chairs, a drafting stool and 3 m of shelves, and equipped with a telephone, which shall be operational throughout the period of fabrication. All long distance telephone calls will be paid for by the Ministry.

The Contractor shall clean, heat and light the office throughout the period of fabrication.

EQUIPMENT

421.21 Qualifications and Equipment

421.21.01 Contractor - The Contractor shall produce evidence that the plant is currently fully approved by the CWB to the requirements of CSA W47.1, Division 1 or Division 2 prior to beginning work.

The Contractor shall also produce evidence of at least five years of satisfactory experience in the fabrication of heavy structural steelwork.

The Contractor shall employ or retain a registered professional engineer experienced in steel bridge fabrication, to provide guidance throughout the work.

A qualified welding supervisor shall be employed on each shift where welding is done on the work.

Prior to fabrication, the Contractor shall submit to the Quality Manager the names of the welding engineer, the engineer experienced in steel bridge fabrication, and the welding supervisors who are to be employed on the work.

421.21.02 Welders and Welding Operators - The Contractor shall produce evidence that all welders and welding operators to be employed on the work are currently qualified by the CWB at the time of fabrication in the processes in which they are to be employed on the work.

The Contractor shall also produce evidence relative to each welder and welding operator, that they have been executing satisfactory welding in the required processes within the six-month period previous to the award of this Contract.

421.21.03 Welding Equipment - All equipment to be used in the work shall be in good working order and shall be subject to the inspection of the Quality Manager.

For any arrangement of automatic welding, the Quality Manager or the Ministry Representative may require that a preliminary test run of the equipment be made, without welding, over the length of the joint, to prove that the disposition of the equipment and the method and accuracy of travel are satisfactory.

421.22 Welding Procedures - The Fabricator shall submit copies of the proposed welding procedures for review by the Quality Manager. Gas metal arc welding shall not be used. Where the submerged arc or flux cored arc process is to be used, the Quality Manager or the Ministry Representative may order that each welder and welding operator make a weld specimen not less than 1 m in length for fillet welds and 150 mm in length for butt welds. Steel of the same specifications and thickness as that to be used in the Work shall be used in the specimen welds.

No welding shall be done on the Work until the welders' and welding operators' qualifications are established to the satisfaction of the Quality Manager and Ministry Representative.

Welding procedures shall be accompanied by documentary proof that they have been qualified previously by the CWB at the plant where the work is to be carried out.

The procedures shall include the following information: joint type, welding process, welding position, base metal

specification, welding consumable specification and size, preheat requirements, amperage and voltage requirements, speed, polarity, and welding equipment, including a description of travel for automatic welding.

421.23 Butt Joints - Except as called for on the Design, butt joints will not be permitted.

The Fabricator may submit an alternative butt joint design for approval by the Designer provided that such design has been approved by the CWB.

421.24 Assembly and Welding Sequences - If requested by the Designer or the Ministry Representative, the Contractor shall supply full details of the proposed assembly and welding sequence of any particular weldment.

CONSTRUCTION

421.31 Preparation of Material

421.31.01 Straightening Material - Prior to being used in fabrication, all structural steel shall be straight and free from kinks or bends. The flatness tolerance of plate in excess of 900 mm wide shall be in accordance with the tolerances of the finished product as specified in CSA W59-M. If straightening is necessary, it shall be done by methods that will not injure the metal. The steel shall not be heated unless permission is given by the Designer and the Ministry Representative.

In no case shall the temperature of the steel exceed 620°C. After straightening, the surfaces of the metal shall be carefully inspected for evidence of fracture and if necessary, the material shall be replaced or repaired to the satisfaction of the Designer and Ministry Representative. Sharp kinks and bends will be cause for rejection of the steel.

421.31.02 Camber - Girders shall be cambered as shown on the Design.

421.31.03 Edge Preparation - Steel may be cut to size by sawing, shearing or flame cutting. All steel before cutting shall be marked by a method agreed to by the Quality Manager and Ministry Representative so that its specification may be immediately identified.

All cut edges shall be smooth and regular, free from fins, cracks, tears and notches. Freehand cutting shall be done only where approved by the Quality Manager and Ministry Representative. Roughness of cut surfaces shall not exceed the U. S. Standards Institute B46.1 value of 1000. Roughness exceeding this value shall be removed by machining or grinding. Occasional gouges will be tolerated only at the discretion of the Quality Manager and Ministry Representative and shall be repaired as instructed.

Sheared edges of plates more than 16 mm in thickness shall be planed to a depth of 6 mm.

Re-entrant flame cuts shall be filleted to a radius of not less than 20 mm.

All exposed sharp corners of members, which are to be painted, shall be ground to a minimum radius of 1.6 mm.

All corners of oxygen-cut edges of main stress-carrying members, except unpainted bearing stiffeners and girder webs, shall be ground to a minimum radius of 1.6 mm.

Special attention shall be given to the cutting of flange plates. All flange plates prepared by flame cutting shall be preheated in accordance with DBSS 421.34.

The Quality Manager is required to carry out Brinell hardness testing of the edges of flange plates on fracture critical members. The Brinell hardness of the edges of flange plates on fracture critical members shall not exceed 220. If greater hardness is measured, the edges shall be ground by the Fabricator to remove the hard layer, or annealed by the Fabricator by means of a preheating torch.

Surfaces to be welded shall be free from loose scale, slag, rust, grease, moisture or other material that will prevent proper welding. Mill scale that withstands vigorous wire brushing, a light film of drying oil or a thin rust inhibitive coating may remain except that all mill scale shall be removed from the surfaces on which flange-to-web welds are to be made. Surfaces within 55 mm of any weld location shall be free from any paint or other material that would prevent proper welding or produce objectionable fumes while welding.

Edges of material thicker than specified in the following list shall be trimmed if and as required to produce a satisfactory welding edge wherever a weld along the edge is to carry calculated stress:

a) Sheared edges of material thicker than 12 mm;

b) Rolled edges of plates (other than Universal Mill Plates) thicker than 10 mm;

c) Toes of angles or rolled shapes (other than wide flange sections) thicker than 16 mm;

d) Universal Mill Plates or edges of flanges of wide flange section thicker than 25 mm.

421.31.04 Direction of Rolling - Steel plates for main members and splice plates for flanges and main tension members shall be cut and fabricated so that the direction of rolling is parallel to the direction of the primary stresses.

421.31.05 Bolt Holes - Standard holes for high tensile bolts shall be either punched, sub-punched and reamed, or drilled, and shall be of a nominal diameter not more than 2 mm in excess of the nominal bolt diameter, except that the following bolt/hole combinations will be permitted:

a) either 3/4 inch or M20 bolts in 22 mm holes;

b) either 7/8 inch or M22 bolts in 24 mm holes;

c) either 1 inch or M24 bolts in 27 mm holes.

Oversize or slotted bolt holes will be allowed only in special circumstances with the approval of the Designer.

Punched holes shall be clean cut, without torn or ragged edges. The diameter of the die shall not exceed the diameter of the punch by more than 2 mm. If a punched hole must be enlarged to admit a high tensile bolt, it shall be reamed.

Reamed holes shall be cylindrical and perpendicular to the member. Where practicable, reamers shall be directed by mechanical means. Reaming shall be done with twist drills or reamers.

Drilling shall be done with twist drills or core drills. Burrs on the outside surfaces shall be removed.

Poor matching of holes will be cause for rejection.

421.31.06 Faying Surfaces - All faying surfaces of steelwork shall be cleaned by sand blasting in the shop. Faying surfaces of steelwork to be painted shall be painted only with one coat of inorganic zinc primer. Primer coatings shall meet the Class B coating requirements as specified in CAN/CSA-S6-00. The class of coating shall be determined based on testing in accordance with the "Specifications for Structural Joints Using ASTM A 325 or A 490 Bolts" issued by the Research Council on Structural Connections. Primers shall be supplied with a Class B certificate and be applied and cured at the conditions on the certificate. Over thickness will not be accepted. If over thickness occurs then the surface will be cleaned and recoated.

421.32 Marking - Prior to fabrication, all steel shall be marked for identification by heat number and specification by a marking system approved by the Quality Manager and Ministry Representative. Steel, which is unidentified, shall not be used in the Work.

421.33 Assembly of Weldments - The shop assembly of the various components of the weldments shall be executed in accordance with CSA W59-M.

Tack welding shall be done by qualified welders, using the smallest size weld required to hold the components of the assembly together. Tack welds shall be incorporated in the final weld.

421.34 Preheat and Interpass Temperatures - No welding shall be done when the temperature of the base metal is lower than -18°C. At temperatures below 0°C, the steel shall be preheated to a temperature of at least 10°C in excess of that stated in CSA W59-M, Table 5.3.

Preheat shall be applied to all steel to be welded so that the steel within 75 mm of the weld is heated to the temperature shown in CSA W59-M, Table 5.3.

Preheat shall be applied in such a manner that moisture from the heating equipment does not penetrate the joint.

Preheat temperatures above the minimum shown in CSA W59-M, Table 5.3 may be required for highly restrained joints if designated by the Designer.

Preheat temperature shall in no case exceed 200°C.

Preheat requirements for tack welds shall be as in the above table except that where single pass tack welds are used and are to be incorporated and consumed in a weld made by the submerged arc process, preheat is unnecessary.

421.35 Welding - Welding shall be done by the manual, shielded metal arc, flux-cored arc or submerged arc processes in accordance with the reviewed procedures and CSA W59-M, Section 5.

Any weld between the web and flange of a bending member shall be made by an automatic process, which provides a continuous weld throughout the length of the member.

All welding shall be done under cover and, in the case of flux-cored arc welding, shall be done in an area free from wind or draft.

Where the submerged arc process is to be used, the Contractor shall:

a) Carry out a preliminary test run of the procedure over the length of the joint to prove that the disposition of the equipment, the handling of hoses, and the method and accuracy of travel are satisfactory.

b) Have each operator make a weld specimen not less than 1200 mm in length for fillet welds and 150 mm in length for butt welds. Steel of the same specification and thickness as that to be used in the work shall be used in the specimen welds. No welding shall be done on the Work until such a specimen is satisfactory to the Quality Manager.

Butt welds shall be extended beyond the edges of the parts to be joined by means of start and run-off tabs providing sufficient thickness to avoid the weld burning through and with a joint preparation similar to that on the main material. For manual shielded metal arc welding the width of the tabs shall be not less than the thickness of the thicker part being joined or 75 mm, whichever is greater. For submerged arc, welding the width of the tabs shall be not less than 75 mm. Each weld pass shall be carried far enough beyond the edge of the parts being joined to ensure sound welds in the joint. Tabs shall be removed upon completion, and the weld cooled without damage to the parent plate. The end of the weld shall be made smooth and flush with the edges of the abutting parts.

In flux-cored arc welding the equipment shall be capable of sustaining a gas flow rate of from 0.85 to 1.25 m^3/h.

421.36 Shear Connector Studs - Shear connector studs shall be welded in the locations shown on the Design, to the requirements of CSA W59-M, Section 5.5.6. Shear connectors will be inspected by the Quality Manager and

shall be repaired if necessary in accordance with the same standard.

421.37 High-Strength Bolts - Installation of high-strength bolts shall be in accordance with the AASHTO Specifications for Highway Bridges (17th Edition - 2002 including all subsequent interim revisions and updates), Division II, Section 11.5.6.

Galvanized bolts shall be lubricated with beeswax or other approved lubricant before installation.

Heads of bolts shall be placed on the outsides of girders and box members, and generally on the more conspicuous side, if any, of any connection.

421.38 Bent Plates - When bending plates, the plates shall be so taken from the stock plates that the bend line will be at right angles to the direction of rolling. The radius of the bend measured inside shall be not less than the thickness of the plate. Before bending, the corners of the plate shall be rounded to a radius of 2 mm throughout that portion of the plate at which bending is to occur.

421.39 Shop Assembly of Bolted Connections - Holes in girder and truss field splices shall be drilled while assembled in the shop or yard, or sub-punched or sub-drilled and reamed while assembled. Unless otherwise specified, the structure shall be progressively assembled in accordance with AASHTO Standard Specifications for Highway Bridges (17th Edition - 2002 including all subsequent interim revisions and updates), Division II, Section 11.5.3.1. The assembly, including camber, alignment and accuracy of holes shall be inspected by the Quality Manager before drilling or reaming is commenced.

Connecting parts assembled in the shop or yard for the purpose of reaming or drilling holes shall be match-marked by a method agreed to by the Quality Manager to indicate the location and orientation of all pieces.

Alternatively, the Contractor may drill holes full-size using automatic drilling equipment, as described in AASHTO Standard Specifications for Highway Bridges (17th Edition - 2002 including all subsequent interim revisions and updates), Division II, Section 11.4.8.3. In this case a check assembly will be required for the first of each major structural type, as described in AASHTO Standard Specifications for Highway Bridges (17th Edition - 2002 including all subsequent interim revisions and updates), Division II, Section 11.5.3.3. Reaming of holes shall be as specified in DBSS 421.47, if the bolt holes do not line up during field assembly.

421.40 Dimensional Tolerances - The dimensions of completed members shall comply with the appropriate dimensional tolerances as specified in CSA W59-M.

The maximum deviation from specified length shall be ± L/1000 but not over 20 mm.

Warpage of box members shall be determined by taking measurements at any two cross-sections in a member, at opposite edges of one face of the member. Warpage is defined as the distance by which any point deviates from a plane defined by the other three points. This warpage shall not exceed 1/200 of the width of the member, or 3 mm, whichever is greater.

421.41 Machined Surfaces - Machine-finished surfaces, as designated on the Design, shall be coated with an approved protective compound.

421.42 Galvanizing or Metallizing (if required) - All steelwork to be galvanized shall be galvanized after complete fabrication to the requirements of CAN/CSA-G164-M. The galvanizer shall safeguard against embrittlement as required in CAN/CSA-G164-M, Appendix A. Galvanized members shall be subject, at the discretion of the Quality Manager or the Ministry Representative, to the tests for embrittlement outlined in CAN/CSA-G164-M, Section 5.5.

The chemical composition of steel being galvanized shall be as follows:

- Carbon less than 0.25%
- Phosphorus less than 0.05%
- Manganese less than 1.35%
- Silicon less than 0.03% or between 0.15% and 0.25%

For steel not meeting these chemical composition requirements, special galvanizing techniques shall be developed by the galvanizer to ensure that the specified coating thickness and adherence is achieved. A detailed description of the special techniques shall be submitted to the Quality Manager and Ministry Representative for review 2 weeks prior to galvanizing.

All steelwork to be metallized shall, after complete fabrication, be treated in accordance with the current CSA G189. The zinc coating shall not be less than 0.3 mm in thickness.

421.43 Shop Painting - The Design and/or the Contract shall specify whether the structure is to be painted, or what parts of a structure are to be painted. This subsection applies to those parts of the steelwork which are to be painted.

For weathering steel bridges, unless specified otherwise in the Contract, all structural steel shall be painted for the larger of the following two distances from deck joint locations:

- 3000 mm; or
- 1.5 times the superstructure depth (including girder, haunch and slab thickness).

Coating work shall be in accordance with DBSS 216 unless noted otherwise in DBSS 421.

All steelwork, which is to be painted, shall be given three shop coats of paint - primer, stripe coat and midcoat. Paint shall be chosen from DBSS 308 System SS1. Paint shall be supplied by the Contractor.

The topcoat may be applied in the shop or in the field. The topcoat coat shall be chosen from either DBSS 308 System SS1 or SF2. If the topcoat is to be applied in the shop, then the Quality Control Program shall specifically address the integrity of the topcoat through to project completion.

Surfaces surrounding bolt holes at connection locations shall receive the prime coat only and shall be masked off so that no stripe coat, midcoat or topcoat paint will be under the bolt heads, washers or nuts. These masked off areas shall have the stripe coat, midcoat and topcoat paint applied in the field after installation of the bolts.

Faying surfaces shall receive the prime coat only, as per DBSS 421.31.06. Faying surfaces shall not be coated with the stripe coat, midcoat or topcoat paint.

Paint shall be applied in a covered area in accordance with the manufacturer's specifications and DBSS 216.

All edges, corners, crevices, bolts, nuts, protrusions and welds (unless ground flush) shall be stripe painted by brush for a width of 50 mm with the midcoat paint before the midcoat coat is applied. Stripe coating paint shall be applied as per SSPC-PA1 and may be applied by spray, but shall be brushed in. Stripe painting shall be allowed to cure before the midcoat coat is applied.

Unless a different coating is called for on the Design, the exposed steel surfaces of bearings shall be cleaned and painted as structural steelwork.

Surfaces shall be cleaned to "near-white" per SSPC-SP10 / NACE No. 2 and 50 μm to 75 μm (2 to 3 mils) sharp/angular profile.

The midcoat shall be applied to all surfaces including tops and sides of the top flanges, except faying surfaces and surfaces surrounding bolt holes. The midcoat shall not be applied until the primer and stripe coat are accepted by the Quality Manager to be sufficiently cured.

Any surfaces inaccessible after erection, except faying surfaces and tops and sides of top flanges, shall be given in addition to the three shop coats, one coat of the topcoat paint appropriate for the paint system being used.

If the topcoat is to be applied in the shop, then the topcoat shall be applied after the midcoat is accepted by the Quality Manager to be sufficiently cured.

The colour of the topcoat will be selected by the Ministry. For weathering steel, unless noted otherwise, the topcoat colour shall match the expected colour of the oxidized surfaces. The proposed colour shall be subject to the acceptance of the Ministry Representative.

421.44 Marking and Shipping - Each member shall be marked by a method agreed to by the Quality Manager and Ministry Representative with an erection mark, corresponding to the mark shown on the erection diagram. Each member with a mass of over one tonne shall also be marked with the mass.

Members shall be loaded on trucks or cars in such a manner that they can be transported to and unloaded at their destination without being damaged.

After steelwork has been delivered to Site it shall be inspected by the Quality Manager. The Contractor shall clean the steelwork after it has arrived at Site of any dirt, road salts, slush or other contaminants accumulated during transport and shall carry out any other surface preparation work necessary to meet the specified surface preparation requirements.

421.45 Field Assembly - The parts shall be accurately assembled as shown on the Design and any matchmarks shall be followed. Hammering which will injure or distort the members shall not be done. Bearing surfaces and surfaces to be in permanent contact shall be cleaned before the members are assembled. Field connections shall have one half of the holes filled with bolts and cylindrical erection pins (half bolts and half pins) before final bolting. Fitting-up bolts shall be the same nominal diameter as the high tensile bolts, and cylindrical erection pins shall be 1 mm larger.

421.46 Straightening Bent Material after Fabrication - The straightening of plates and angles or other shapes shall be done by methods that will not produce fracture or other injury. The metal shall not be heated unless permitted by the Designer and the Ministry Representative, in which case the heating shall not be to a higher temperature than that producing a "dark cherry red" colour. After heating, the metal shall be cooled as slowly as possible.

Following the straightening of a bend or buckle, the surface of the metal shall be carefully inspected for evidence of fracture, and if necessary, replaced or repaired to the satisfaction of the Designer and the Ministry Representative.

421.47 Misfits - For all primary connections, and secondary connections having eight or more bolts, 85% of the holes shall accept bolts without reaming. The remaining 15% may be reamed to accept the designed diameter bolts. The diameter of the reamer shall be the same as the drilled hole.

Holes in plates showing more than 5 mm of offset shall be cause for rejection of that plate. New plates, if necessary, shall be field drilled using the hole pattern in the senior member as a template.

421.48 Erection Tolerances - Unless otherwise specified, the misalignment of members after erection shall be within the dimensional tolerances specified in CSA W59-M, but not over 50 mm.

Misalignment shall be measured from vertical lines in the case of columns or towers, and from lines joining the ends of any test length of a member.

Joints, which are required on the Design to be milled to bear, shall have at least 75% of the entire contact area in full bearing. The separations of any remaining portions shall not exceed 0.25 mm except locally at toes of flanges where a separation of 0.60 mm is permissible.

Where joints are not milled, the opening shall not exceed 13 mm.

421.49 Field Painting - This section applies to those parts of the steelwork which are to be painted. Coating work shall be in accordance with DBSS 216 unless noted otherwise in DBSS 421. After the completion of all deck and overhead concrete work, steelwork shall be thoroughly cleaned of all rust, dirt, dust, oil and other foreign materials. Non-visible salts shall meet the SSPC-SP12 / NACE No. 5 NV-2 levels. The shop coats of paint shall be touched up as necessary.

421.49.01 Field Touch-up - Bare, rusty or damaged areas shall be cleaned to SSPC-SP11, Power Tool Cleaning to Bare Metal. Feather edges into the existing coating and build the coating as per DBSS 216. Build the coating using coatings from the specified DBSS 308 System. The coating system shall be from the same manufacturer as the shop coating system on the steel.

After installation, Type 3 bolts, nuts and washers in the areas to be painted shall be cleaned to SSPC-SP11, Power Tool Cleaning to Bare Metal. These bolts, nuts and washers shall then receive the prime coat, stripe coat, midcoat and topcoat paint.

The masked off surfaces surrounding the bolt holes at connection locations shall be cleaned and shall receive the stripe coat, midcoat and topcoat paint.

421.49.02 Topcoat - The topcoat coat shall be chosen from either DBSS 308 System SS1 or SF2.

If the topcoat has not been applied in the shop, then the steelwork shall be given one topcoat coat of paint applied in accordance with the manufacturer's specifications and DBSS 216.

The colour of the finish coat will be selected by the Ministry. For weathering steel, unless noted otherwise, the topcoat colour shall match the expected colour of the oxidized surfaces. The proposed colour shall be subject to the acceptance of the Ministry Representative.

At the completion of the contract, all steelwork, painted or unpainted, shall be cleaned of concrete spatter, mud, oil and other foreign materials.

421.50 Unpainted Weathering Steel - In the case of unpainted weathering steel, the outer faces of the girders and stringers and any other surfaces mentioned in the Design shall present a uniform surface free of mill scale and, if necessary, shall be power washed and/or sandblasted prior to installation. Concrete splatter adhering to the steel surfaces after the construction of the deck shall be removed and the steel surfaces cleaned.

421.51 Touch-up of Galvanizing and Metallizing - All field welds and other damage in galvanized and metallized coatings shall be touched up as follows:

If the Design calls for touch-up by metallizing, the damaged areas shall be locally sandblasted to "near white" per SSPC-SP10 / NACE No. 2. All dry abrasive blast cleaned areas shall be metallized in accordance with CSA G189, to provide a zinc coating not less than 0.3 mm in thickness.

Otherwise, the damaged areas shall be thoroughly cleaned and painted with two coats of Ministry-approved organic zinc-rich paint.

QUALITY CONTROL PROGRAM

421.61 Quality Control Program - These Subsections describe the general Quality Control Program required for any Fabricator undertaking the fabrication of permanent steel bridges and steel bridge components. The Quality Control Program shall be part of the overall Quality Management Plan for the project.The term "Quality Control" defines those activities that the Fabricator performs to conform to the Contract.

The term "Quality Assurance" defines those activities that the Quality Manager performs to ensure conformance to the Contract.

The term “Quality Audit” defines those activities that the Ministry performs to provide confidence that the QC and QA processes and resulting products satisfy the Contract. .

These Subsections contain statements of the Quality Objectives and Policies that the Ministry considers essential for successful and economical quality management. They also outline the Procedures and Documentation to implement and confirm that objectives are met.

The provisions of the Quality Control Program set forth in these Subsections shall apply to all steel bridges and bridge components required as part of the Work.

These Subsections require the establishment of a Quality Management Team with the responsibility for the successful and timely implementation of all necessary Quality Control and Quality Assurance activities. Some positions shown in the Suggested Organization Chart may be held by the same individual. For example, the General Manager may also be the Contract Administrative Manager and the Purchasing Manager. The Plant Superintendent may also be the Receiver and the Welding Supervisor.

421.62 Quality Objectives and Policies

421.62.01 Quality Objectives - The Quality Objectives of the Ministry cover all steel bridges and steel bridge components produced under this Contract to the Ministry, as follows:

a) Completed products shall conform fully to the governing Codes and Specifications stipulated in the Design and the Contract.

b) The Quality Control Program shall be fully integrated into the ongoing manufacturing activities of the Fabricator.

421.62.02 Quality Policy - The Quality Policy ensures that the product meets the Quality requirements of the Contract and is delivered on time.

The Quality Control Program applies to all stages of the design, drafting, procurement, manufacturing and testing of the product.

A Fabricator's Quality Control Manager shall be appointed with defined responsibilities in resolving quality matters and shall report to a senior management level. At each hold point, the Fabricator's Quality Control Manager shall:

- Document the successful completion of each stage as it progresses through fabrication and erection;
- Identify and report nonconforming components;
- Initiate or recommend disposition of nonconforming components;
- Verify corrections.

Any persons assigned to perform quality control inspections shall be other than those performing or directly supervising the work and they shall not report directly to immediate supervisors responsible for producing the work.

The Quality Control Program is not subordinate to any design, drafting, procurement, manufacturing or testing activities.

421.63 Scope of the Quality Control Program - The Quality Control Program governs the fabrication of steel bridges and bridge components for the Work. The Suggested Organization Chart is shown in Figure 1

The Fabricator's General Manager shall be responsible for:

- adhering to the Quality Control Program in all respects.
- ensuring that completed bridges or bridge components shall conform fully to the Design and to the fabrication and welding requirements stipulated in the Contract.
- ensuring that all required documentation is produced according to the Quality Control Program.

421.64 Range of Capability - The Fabricator shall provide the necessary knowledge, skill (in-house or on a contract basis) and equipment to perform the following work on steel bridges and bridge components:

- Design of connections and joints not shown on Design;
- Preparation of shop fabrication drawings;
- Preparation of bills of material;
- Preparation of material requisitions and purchase orders;
- Receiving, checking and storing materials for bridges;
- Layout, cutting, forming and fitting of parts;
- Assembly, tacking and welding;
- Dimensional checking and verification;
- Resolution of non-conformances;
- Documentation of all stages of work with capability of tracking all major components;
- Cleaning, painting, storing and shipping;
- Erection of bridges and bridge components .

421.65 Drawings and Specifications - The Fabricator's Chief Draftsperson shall:

- obtain the latest revision of the Design and specifications for the Work;
- submit shop drawings and erection drawings to the Designer and Ministry Representative before commencement of the work;
- prepare material requisitions containing a full description of the material sizes, material specifications, and certifications required for conformance to the contract;
- deliver the material requisitions to the Fabricator's Purchasing Manager in ample time to permit ordering, delivery and documentation without delaying the work;
- issue requisitions for all sub-contracted drafting work and shall ensure that all conditions of the Contract are part of such sub-contracts.

The Fabricator's Chief Design Engineer shall be responsible for the design of any connections or joints not shown on the Design; and for the design of the erection procedures and any special erection equipment needed, if required.

421.66 Material Control - All materials for bridges and bridge components shall be ordered by the Fabricator's Purchasing Manager in full conformance with the material requisitions provided by the Fabricator's Chief Draftsperson.

The Purchase Orders shall contain all information

necessary to ensure that materials purchased will comply fully with the terms of the Contract. Where mill certificates and test reports are required, it shall be so stated on the Purchase Order. Instructions shall state when the certificates and reports are to be delivered to the Fabricator.

If a supplier proposes a substitute for any material, the Fabricator's Purchasing Manager shall refer the proposed substitution to the Fabricator's Chief Design Engineer for review. If the substitute is acceptable to the Designer and the Ministry Representative, the Fabricator's Chief Draftsperson shall amend all drawings and requisitions, withdraw old issues, and issue the new versions.

The Fabricator's Receiver shall:

- inspect all materials on arrival for conformance with the Purchase Orders;
- confirm that mill certificates and test reports are provided and that they correctly identify the materials delivered;
- arrange with the Fabricator's Plant Superintendent to store all materials for the Contract in segregated areas. Clear identification with the Contract shall be provided;
- issue a non-conformance report covering overage, shortage or damage to the materials, and copies of the report shall be provided to the Fabricator's Purchasing Manager and the Fabricator's Quality Control Manager.

The Fabricator's Purchasing Manager shall deliver all documentation to the Fabricator's Quality Control Manager for inclusion in the Quality Control file for the Contract.

421.67 In-Progress Inspection and Reporting - The Fabricator's Quality Control Manager shall ensure that only documented materials are used for the Contract.

All materials intended for incorporation into the Work shall be examined after cutting to size, forming and rolling. The Fabricator's Chief Inspector shall ensure conformance with the detailed shop drawings, shall report any non-conformance to the Fabricator's Quality Control Manager and shall order all work affected by the non-conformance to stop, pending approval of remedial action.

Before assembling any plate girders or complex parts, the surfaces of all materials shall be examined for imperfections revealed during previous fabrication operations. The joint edge preparation for all groove welds shall be verified as conforming to the Fabricator's CWB-approved welding standards and shall be within the acceptable tolerances.

After assembly of any plate girders or complex parts, and before commencing the strength welding, the assembly shall be checked for dimensional conformance. The Fabricator's Welding Supervisor shall ensure that the fit-up of all welded joints conforms to the approved welding standards.

The Fabricator's Quality Control Manager shall file a written report of verification with the Fabricator's General Manager and report any non-conformance.

421.68 Correction of Non-conformance When a non-conformance is encountered, the Fabricator's Quality Control Manager shall determine a recommended disposition and obtain the Quality Manager and Ministry Representative's approval as quickly as possible.

If there is non-conformance to the material specification stipulated in the Purchase Order, the Fabricator's Purchasing Manager shall immediately find out the reasons for the delivery of non-conforming material. If the material is of a grade superior to that ordered, the Fabricator's Quality Control Manager and the Fabricator's Chief Engineer shall be notified. They shall verify that the material is an acceptable alternative in all respects, and this verification shall include consultation with the Quality Manager and Ministry Representative. If the material is of a grade inferior to that ordered, it shall be rejected and the correct material or a superior material shall be obtained.

If there is non-conformance of material delivery that will delay production, the Fabricator's Contract Administration Manager and the Fabricator's Plant Superintendent shall be notified immediately. They shall be given revised delivery dates for the adjustment of production scheduling. The Fabricator's General Manager shall determine the alternatives available and shall notify the Quality Manager and the Ministry Representative.

If there are non-conformities in material preparation, assembly, joint edge preparation and fit-up before strength welding, the Fabricator's Quality Control Manager and the Fabricator's Welding Supervisor shall immediately review the non-conformance and notify the Fabricator's Chief Design Engineer who may require further investigation prior to submitting corrective action to the Quality Manager and Ministry Representative for approval. If the necessary corrective action will result in delay to production, the Fabricator's General Manager shall be notified for adjustment to the production schedule. The Fabricator's Plant Superintendent shall inform the Fabricator's Quality Control Manager when the corrective actions are being done so that conformance can be verified and the non-conformance report cancelled.

421.69 Welding - All welding on structural and mechanical components shall be done by the company certified to CSA W47.1 (Division 1 or 2),W47.2-M (Division 2.1 or better) and W186-M as applicable, and shall be done in accordance with the Fabricator's CWB-approved welding standards.

The edge preparations for all groove welds shall conform to the dimensions established in the approved welding

standards.

If the joint is to be welded from one side only without back-gouging, the root gap and root face shall be checked to ensure conformance with the required geometry so that the root pass can be successfully completed.

The strength level and chemical composition of all filler materials used in structural and mechanical assemblies shall conform to the approved shop drawings.

All welding consumables shall conform to the approved welding standards and shall be received, stored and conditioned according to the applicable welding standards.

Any preheat required before welding shall be according to the approved welding standards.

The welding procedure followed in welding any joint in a structural or mechanical component shall conform to the applicable Data Sheet in the approved welding standards.

All welders and welding operators welding on structural or mechanical contracts shall be qualified under the requirements of the CSA Standard governing certification.

421.70 Heat Treatment - This covers any post-weld heat treatment necessary to conform to the approved welding standards.

The Fabricator's Chief Design Engineer shall:

- determine whether any structural or mechanical components shall receive post-weld heat treatment to conform to the Design or Contract or to the applicable Codes and Standards.
- inform the Fabricator's Chief Draftsperson of any such requirements so that they may be incorporated on to the approved shop drawings.

Any heat treatment stipulated on the approved shop drawings or contained in the approved welding standards shall be carried out at the appropriate time and according to the approved documents.

The Fabricator's Plant Superintendent shall ensure that any heat treatment stipulated is done according to the established procedures, shall obtain all documentation and reports, and shall deliver them to the Fabricator's Quality Control Manager.

421.71 Non-destructive Testing - The Quality Manager will test welds by non-destructive testing methods, as described in DBSS 421.06. The Ministry may test any weld by non-destructive testing methods, as described in DBSS 421.06

The Fabricator's Plant Superintendent shall schedule the manufacturing operation to facilitate non-destructive testing.

If non-destructive testing of welded joints reveals imperfections that are marginally more than the acceptance standards, the Fabricator's Quality Control Manager shall consult with the Designer and the Ministry Representative regarding the location and nature of the imperfections. The effects of leaving minor defects in place shall be assessed in relationship to the loads carried by the joint and the possible adverse effects of making an unnecessary repair.

Weld repairs shall conform to the approved welding standards.

421.72 Calibration of Measurement and Test Equipment - All measurement and testing equipment owned by the Fabricator and used in the Quality Control Program shall be calibrated and re-calibrated at the intervals and in the manner stipulated in the Manufacturer's Instruction Manuals. Any adjusting devices shall be sealed or otherwise protected from unauthorized adjustment or tampering.

421.73 Records Retention - The Fabricator's record file for each Contract shall contain the pertinent drawings, purchase orders, bills of material, material mill certificates, test reports, Quality Control documents, NDT reports and certificates of compliance.

The Fabricator's record file shall be made available to the Quality Manager and the Ministry Representative upon request.

Items in the Fabricator's record file shall be retained as per company policy.

The Fabricator's Quality Control Manager shall ensure that each file is complete in all respects before it is placed in the Company archives.

421.74 Hold Points - In planning the work flow, the Fabricator shall coordinate with the Quality Manager and Ministry Representative to decide "hold" points for inspection or non-destructive testing. A list of hold points shall be drawn up by the Fabricator, Quality Manager and Ministry Representative at a prefabrication meeting. The Quality Manager and Ministry Representative shall be informed of progress so that delays are minimized.

The "hold" points will typically include some of, but not be limited to, the following:

- verification of materials
- after plate is prepared for splicing
- after splicing of plates
- after web to flange weld
- after stiffeners are applied
- camber of girders
- shop assembly
- cleaning and coating
- bearing plate attachment

- shipping arrangements
- subassemblies.

Work shall not proceed past a "hold" point until it has been signed off by Quality Control and Quality Assurance. Reports shall be completed promptly.

421.75 Transportation and Installation Procedures - Transportation and installation procedures shall be prepared and submitted for review before any installation takes place on Site. The procedures shall be sealed by a professional engineer experienced in bridge erection and registered with the APEGBC. Consideration shall be given to the following items during the preparation of the procedures:

a) Girder Transportation

i) Brief description of hauling equipment;

ii) Location of girder support points;

iii) Engineering backup if supports vary from the Design.

iv) Details of coating protection during loading transporting and erection.

b) Installation Drawings

i) Bridge site plan showing piers, abutments and access roads;

ii) Crane make, crane chart, boom length(s) and crane locations;

iii) Mass of girder and access to crane(s);

iv) Special installation equipment such as a launching truss, head frames and falsework.

c) Commentary

i) Brief point form description of installation sequence.

d) Bearings

i) Placement and temperature setting procedure for bearings to be included for multiple span continuous girders.

e) Traffic Control

i) Any arrangements that will be made for road and/or rail traffic.

f) Utilities

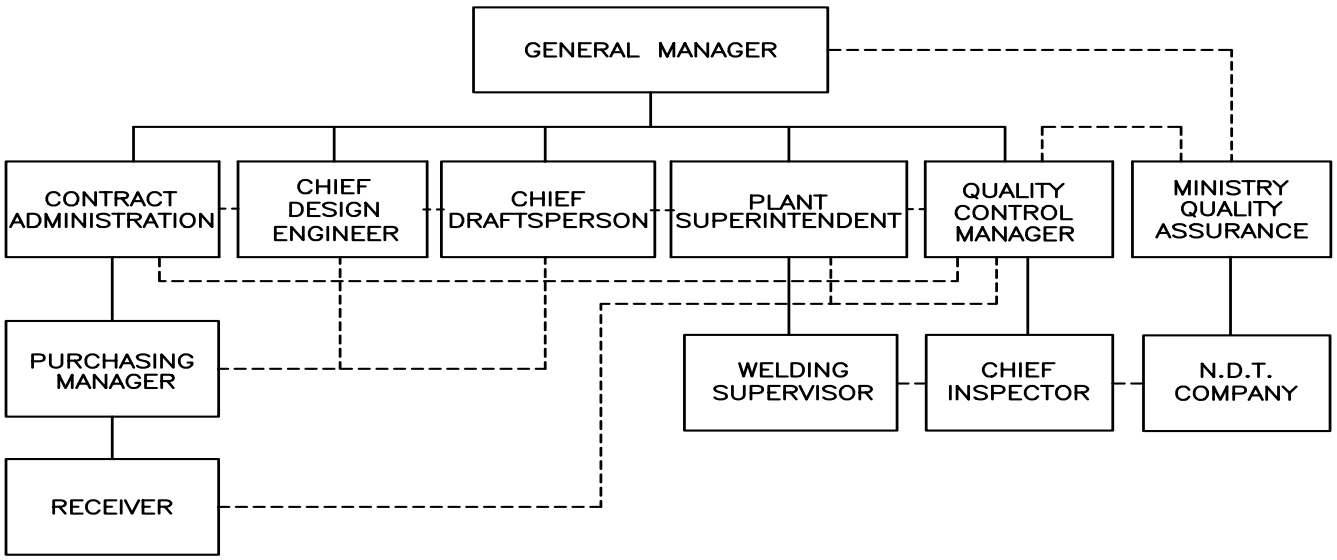
i) Safety and protection.

g) Fall Protection

i) Method and date of installation as required in specifications.

Figure 1

SUGGESTED ORANIZATION CHART



Direct Functional Reporting _____

Quality Control Reporting -

Signed _____
General Manager

Date _____

SAMPLE FORM #1 - RECEIVING REPORT

RECEIVING REPORT

DATE: _____

SUPPLIER _____ P.O. No. _____ WORK ORDER No. _____ INVOICE No. _____

QUANTITY DESCRIPTION MILL CERTS. HEAT No. VISUAL INSPECTION THICKNESS CHECK

HOLD MATERIAL: Describe material, problems and procedures.

QUALITY CONTROL APPROVAL: _____

SAMPLE FORM #2

NON-CONFORMANCE REPORT